

## SCHLESWIG JAGEL, Germany

WMO: 100370

Lat: 54.4569N

Lon: 9.5203E

Elev: 82

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 15.7       | 19.8 | 11.8                          | 10.1 | 18.6 | 15.7 | 12.2 | 22.4 | 35.5                  | 46.3 | 32.4 | 44.8 | 6.2                      | 120  | 0.634 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.6                   | 82.5            | 66.5 | 78.7 | 65.2 | 75.1 | 63.5 | 68.3                | 78.9 | 66.4 | 75.9 | 64.6 | 72.8 | 10.0                 | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.4                            | 90.9 | 72.7 | 62.7 | 85.4 | 71.0 | 61.0 | 80.4 | 68.6 | 32.5          | 79.3 | 31.0 | 75.9 | 29.7 | 73.1 | 77.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 28.2          | 24.6 | 21.5 | DB                         | 8.6 | 89.2               | 6.0 | 3.3                                                | 4.3 | 91.5       | 0.8 | 93.5       | -2.6 | 95.3       | -6.9 | 97.7 | (4) |
| (5)               |      |      | WB                         | 8.4 | 72.2               | 5.8 | 1.7                                                | 4.2 | 73.4       | 0.8 | 74.4       | -2.4 | 75.4       | -6.6 | 76.6 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   |         | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------|-------------------------------------------------------------------|---------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   |         | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg   | 48.3        | 34.3  | 35.0  | 39.1  | 46.4  | 53.4  | 59.0  | 63.7  | 63.2  | 57.3  | 49.6  | 41.6  | 36.3  | (6)  |
| (7)  |                                                                   | DBStd   | 12.22       | 7.51  | 7.24  | 6.31  | 6.23  | 5.79  | 5.28  | 5.54  | 5.08  | 4.83  | 5.90  | 6.26  | 7.65  | (7)  |
| (8)  |                                                                   | HDD50   | 2186        | 488   | 419   | 339   | 143   | 30    | 1     | 0     | 0     | 3     | 79    | 259   | 425   | (8)  |
| (9)  |                                                                   | HDD65   | 6204        | 953   | 839   | 802   | 557   | 361   | 195   | 95    | 97    | 236   | 476   | 703   | 890   | (9)  |
| (10) |                                                                   | CDD50   | 1571        | 0     | 0     | 2     | 35    | 136   | 270   | 423   | 410   | 221   | 67    | 6     | 1     | (10) |
| (11) |                                                                   | CDD65   | 116         | 0     | 0     | 0     | 0     | 2     | 14    | 54    | 42    | 4     | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74   | 981         | 0     | 0     | 0     | 5     | 26    | 139   | 456   | 322   | 32    | 1     | 0     | 0     | (12) |
| (13) |                                                                   | CDH80   | 221         | 0     | 0     | 0     | 0     | 4     | 29    | 112   | 74    | 2     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg   | 10.1        | 11.3  | 11.7  | 11.7  | 10.4  | 9.9   | 9.6   | 9.1   | 8.4   | 8.8   | 9.5   | 9.8   | 10.7  | (14) |
| (15) | Precipitation                                                     | PrecAvg | 35.7        | 3.2   | 2.1   | 2.2   | 1.9   | 2.3   | 2.9   | 3.5   | 3.6   | 3.3   | 3.5   | 3.7   | 3.5   | (15) |
| (16) |                                                                   | PrecMax | 46.9        | 7.0   | 5.7   | 5.4   | 5.5   | 5.2   | 6.2   | 7.3   | 9.8   | 7.4   | 7.6   | 10.9  | 8.7   | (16) |
| (17) |                                                                   | PrecMin | 20.9        | 0.0   | 0.4   | 0.3   | 0.1   | 0.2   | 0.1   | 0.6   | 0.1   | 0.8   | 0.6   | 0.7   | 0.6   | (17) |
| (18) |                                                                   | PrecStd | 5.7         | 1.5   | 1.1   | 1.2   | 1.1   | 1.2   | 1.5   | 1.7   | 2.0   | 1.6   | 1.6   | 2.0   | 1.8   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%    | DB          | 51.4  | 52.3  | 60.8  | 72.5  | 77.1  | 84.5  | 88.0  | 86.5  | 78.6  | 67.7  | 57.7  | 53.4  | (19) |
| (20) |                                                                   | 0.4%    | MCWB        | 48.6  | 46.9  | 50.5  | 57.8  | 61.6  | 66.4  | 68.1  | 67.3  | 65.9  | 60.4  | 54.5  | 51.0  | (20) |
| (21) |                                                                   | 2%      | DB          | 48.3  | 48.8  | 54.5  | 66.0  | 72.0  | 78.0  | 82.8  | 81.9  | 73.1  | 62.7  | 54.9  | 50.6  | (21) |
| (22) |                                                                   | 2%      | MCWB        | 46.2  | 45.7  | 47.6  | 53.8  | 59.5  | 64.2  | 66.5  | 66.5  | 62.7  | 57.4  | 52.2  | 48.5  | (22) |
| (23) |                                                                   | 5%      | DB          | 46.3  | 46.6  | 50.6  | 61.5  | 68.4  | 73.3  | 79.1  | 77.3  | 68.7  | 60.2  | 52.3  | 48.2  | (23) |
| (24) |                                                                   | 5%      | MCWB        | 44.4  | 44.3  | 46.0  | 52.1  | 57.3  | 61.9  | 65.3  | 65.2  | 60.7  | 56.1  | 49.6  | 46.6  | (24) |
| (25) |                                                                   | 10%     | DB          | 44.3  | 44.6  | 48.2  | 57.3  | 64.7  | 69.6  | 75.1  | 73.4  | 65.7  | 58.1  | 50.2  | 46.3  | (25) |
| (26) |                                                                   | 10%     | MCWB        | 42.7  | 42.4  | 44.6  | 50.2  | 55.5  | 60.1  | 63.8  | 63.5  | 59.0  | 54.7  | 48.0  | 44.8  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%    | WB          | 49.0  | 48.5  | 52.1  | 58.7  | 64.4  | 69.4  | 70.4  | 70.4  | 66.8  | 61.1  | 55.8  | 51.4  | (27) |
| (28) |                                                                   | 0.4%    | MCDWB       | 50.6  | 51.0  | 58.1  | 69.0  | 73.5  | 81.0  | 81.7  | 80.5  | 75.5  | 66.2  | 57.9  | 53.4  | (28) |
| (29) |                                                                   | 2%      | WB          | 46.3  | 45.9  | 48.9  | 55.4  | 61.2  | 65.9  | 68.4  | 67.9  | 63.7  | 58.5  | 52.6  | 48.8  | (29) |
| (30) |                                                                   | 2%      | MCDWB       | 47.8  | 47.8  | 52.9  | 63.5  | 69.4  | 75.6  | 79.7  | 77.7  | 70.5  | 61.9  | 54.8  | 50.2  | (30) |
| (31) |                                                                   | 5%      | WB          | 44.3  | 44.3  | 46.8  | 52.9  | 58.8  | 63.2  | 66.6  | 66.0  | 61.8  | 56.7  | 50.2  | 46.5  | (31) |
| (32) |                                                                   | 5%      | MCDWB       | 45.6  | 46.2  | 50.2  | 60.0  | 65.8  | 70.6  | 76.7  | 74.8  | 67.3  | 59.7  | 52.0  | 47.8  | (32) |
| (33) |                                                                   | 10%     | WB          | 42.4  | 42.5  | 44.7  | 50.6  | 56.6  | 60.9  | 64.6  | 64.3  | 60.0  | 55.0  | 48.2  | 44.6  | (33) |
| (34) |                                                                   | 10%     | MCDWB       | 43.8  | 44.4  | 47.4  | 56.5  | 63.4  | 67.8  | 73.3  | 71.6  | 64.7  | 57.9  | 49.9  | 45.9  | (34) |
| (35) | Mean Daily Temperature Range                                      |         | MDBR        | 8.0   | 9.3   | 12.4  | 16.2  | 16.8  | 16.2  | 16.6  | 16.6  | 14.5  | 11.9  | 9.2   | 8.1   | (35) |
| (36) |                                                                   | 5% DB   | MCDBR       | 8.1   | 9.9   | 16.8  | 23.0  | 23.4  | 23.1  | 23.3  | 22.7  | 19.1  | 13.8  | 9.5   | 7.8   | (36) |
| (37) |                                                                   | 5% DB   | MCWBR       | 7.6   | 7.9   | 11.3  | 13.2  | 12.9  | 12.2  | 10.4  | 10.7  | 10.9  | 9.8   | 8.1   | 7.3   | (37) |
| (38) |                                                                   | 5% WB   | MCDBR       | 8.0   | 9.4   | 15.0  | 20.3  | 19.4  | 20.3  | 21.2  | 19.5  | 16.7  | 12.3  | 8.9   | 7.8   | (38) |
| (39) |                                                                   | 5% WB   | MCWBR       | 7.7   | 7.9   | 10.8  | 12.3  | 11.8  | 11.9  | 10.4  | 10.4  | 10.3  | 9.3   | 7.9   | 7.5   | (39) |
| (40) | Clear-Sky Solar Irradiance                                        |         | taub        | 0.322 | 0.341 | 0.368 | 0.405 | 0.397 | 0.394 | 0.412 | 0.409 | 0.385 | 0.359 | 0.336 | 0.311 | (40) |
| (41) |                                                                   |         | taud        | 2.287 | 2.309 | 2.265 | 2.212 | 2.294 | 2.327 | 2.296 | 2.316 | 2.364 | 2.397 | 2.342 | 2.282 | (41) |
| (42) |                                                                   |         | Ebn at Noon | 194   | 234   | 253   | 259   | 269   | 271   | 263   | 256   | 246   | 225   | 188   | 171   | (42) |
| (43) |                                                                   |         | Edh at Noon | 19    | 25    | 33    | 40    | 38    | 38    | 38    | 36    | 30    | 23    | 18    | 15    | (43) |
| (44) | All-Sky Solar Radiation                                           |         | RadAvg      | 163   | 366   | 774   | 1295  | 1640  | 1701  | 1601  | 1301  | 928   | 499   | 204   | 118   | (44) |
| (45) |                                                                   |         | RadStd      | 21    | 54    | 69    | 144   | 163   | 141   | 189   | 100   | 102   | 57    | 24    | 9     | (45) |

## Historical Trends

|      |                         | Heating      |            | Cooling    |            |            | Degree-Days |            |             |            |            |      |
|------|-------------------------|--------------|------------|------------|------------|------------|-------------|------------|-------------|------------|------------|------|
|      |                         | 99% DB       | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD50       | HDD65      | CDD50       | CDD65      |            |      |
| (46) | Station Only            | (d)<br>+0.77 | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A  | (j)<br>N/A | (k)<br>-347 | (l)<br>N/A | (m)<br>N/A | (46) |
| (47) | Regional (50 neighbors) | N/A          | N/A        | N/A        | N/A        | N/A        | N/A         | N/A        | -257        | N/A        | N/A        | (47) |

Nomenclature: See separate page